23. A New Chromosome Number in the Indian Spiny Mouse, *Mus platythrix*[*]

By Prasanna KUMARI,[**] Nimmakayalu V. ASWATHANARAYANA,[**] and Tosihide H. YOSIDA[***]

(Communicated by Sajiro MAKINO, M. J. A., April 12, 1984)

The Indian spiny mouse, *Mus platythrix*, has a characteristic karyotype consisting of 26 acrocentric chromosomes with a few larger autosomes (Satya-Prakash and Aswathanarayana 1972; Tsuchiya and Yosida 1972; Yosida 1979; Aswathanarayana and Manjunatha 1982). Yosida (1979, 1980a) derived the karyotype of this species from that of the house mouse, *M. musculus*, by proposing tandem fusions of several pairs of the autosomes. He also suggested that the chromosome evolution might have occurred sequentially by the fusion (Yosida 1983). As predicted by him, Aswathanarayana and Manjunatha (1982) have discovered a female of this species with 2n=25 with a fused metacentric chromosome. During a series of studies we discovered a male with a reduced chromosome number of 24, characterized by all acrocentrics. Since this observation is of great cytogenetic significance, a preliminary account is presented in this communication.

Material and methods. A total of 27 specimens of the Indian spiny mouse (15 females and 12 males) were collected in the environs of a village near Mysore, South India. Bone marrow and spleen cells were utilized for the chromosome observations by the routine air drying technique.

Results and discussion. Among 27 specimens collected in Mysore, 26 showed the usual diploid number (2n=26) with all the acrocentric chromosomes as already described by Satya-Prakash and Aswathanarayana (1972), Tsuchiya and Yosida (1972) and Yosida (1979) (Fig. 1, A and C). One male, however, was remarkable by having a distinct count of 24 chromosomes. All chromosomes were characterized by acrocentrics ranging from large to small in size (Fig. 1, B and D). The same chromosome number (2n=24) was counted in all bone marrow and spleen cells examined so far. Among 12 autosome pairs (nos. 1 to 12) in the original karyotype, one of the

---

[*] Contribution no. 1561 from the National Institute of Genetics, Japan.

[**] Department of Zoology, University of Mysore, Manasa Gangotri, Mysore, India.

[***] National Institute of Genetics, Misima, Japan.

smaller pairs, probably no. 11 appears to be missing in the new karyotype. This pair might have been translocated tandemly to any one of the other autosome pairs as in the other large acrocentrics developed by tandem fusion (Yosida 1980), which will be demonstrated by the banding analysis. Interesting is the fact that the Y chromosome in this specimen was considerably smaller than the other specimens with 2n=26. It seems that in the course of development of 24 chromosome complement, the deletion in the Y element would have taken place.

Variations of the chromosome numbers involving Robertsonian fusions are not unusual in the eutherian mammals, especially in rodents. In the karyotypic evolution of the genus *Mus*, there are many instances of the Robertsonian fusion mechanism (Gropp *et al.* 1972; Capanna *et al.* 1973; Brooker 1982). In all these cases the fundamental number of chromosomes (FN) remains the same, i.e.,

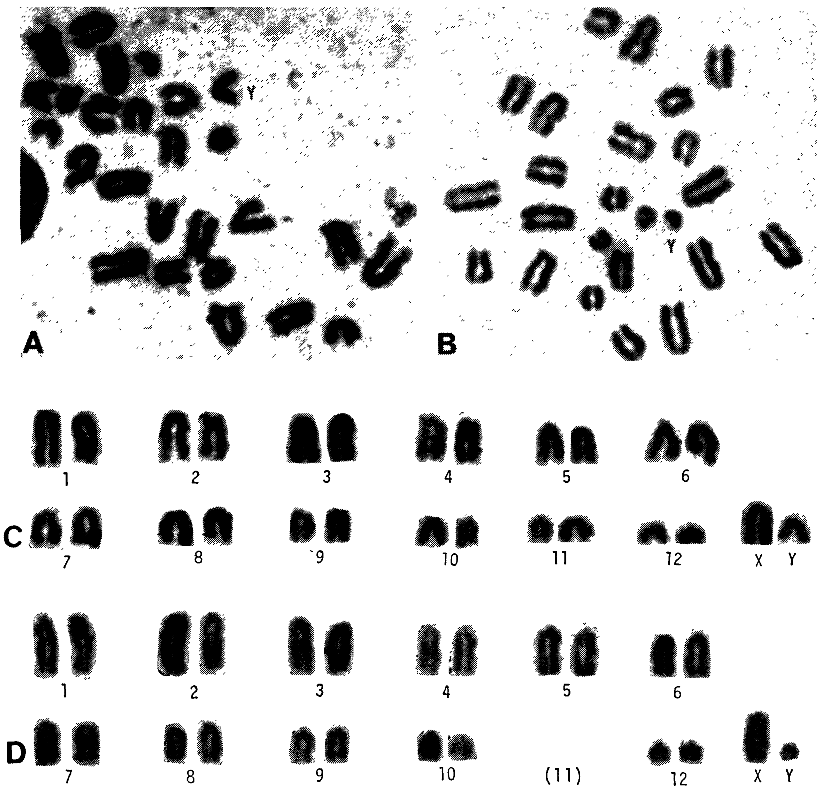

Fig. 1. Metaphases (A and B) and karyotypes (C and D) of the Indian spiny mouse with 26 and 24 chromosomes. A and C, cell with 26 chromosomes. B and D, cell with 24 chromosomes. In the latter cell, no. 11 chromosomes shown by the parenthesis are lost from the karyotype.

40. In the Indian spiny mouse, however, the fundamental number has changed from 26 to 24 by the tandem fusion. Such a change is not only novel but interesting in the cytogenetics of the genus *Mus*. This instance of the karyotype evolution is of great interest, because individuals of the Indian spiny mouse with 24 chromosomes may be found not only in the same population but also in other populations of India. As predicted by Yosida (1983), the karyotype evolution in the Indian spiny mouse would proceed sequentially, and specimens with fewer chromosome numbers other than 24 is possible by a few more tandem fusions.

Acknowledgements. The present authors are grateful to Professor Emeritus Sajiro Makino, M. J. A., for reading of this manuscript with keen interest. They also thank Professor N. B. Krishnamurthy, Department of Zoology, University of Mysore, for his encouragement to this work. This work is supported partially by a research grant from U. G. C., New Delhi, India (N. V. A.) and by a grant-in-aid from the Ministry of Education, Science and Culture of Japan (T. H. Y.).

References

Aswathanarayana, N. V., and Manjunatha, K. R.: Mamm. Chrom. Newsl., **22**, 57–58 (1981).
Brooker, P. C.: Heredity, **48**, 305–309 (1982).
Capanna, E. *et al.*: Chromosoma (Berl.), **58**, 341–353 (1976).
Gropp, A., Tettenborn, U., and Lehmann, E.: Cytogenetics, **9**, 9–23 (1970).
Satya-Prakash, K. L., and Aswathanarayana, N. V.: Mamm. Chrom. Newsl., **13**, 120 (1972).
Tsuchiya, K., and Yosida, T. H.: Ann. Rep. Nat. Inst. Genet. Japan, **22**, 51–52 (1972).
Yosida, T. H.: Proc. Japan Acad., **55**B, 270–274 (1979).
——: Cytologia, **45**, 753–762 (1980).
——: Proc. Japan Acad., **59**B, 5–8 (1983).